Prof. Hussein MA

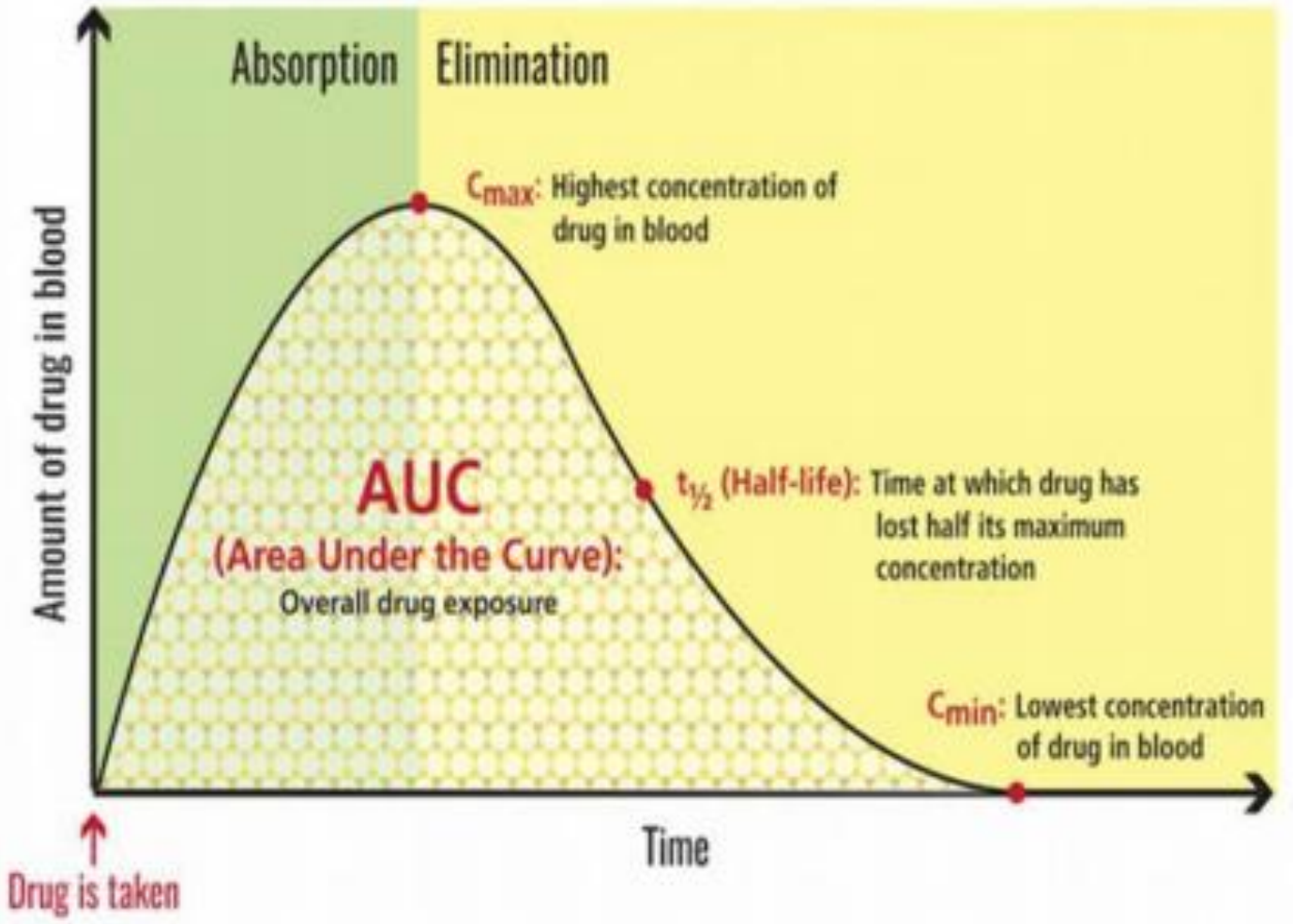
Pharmacokinetics

Pharmacokinetics is what your body does to a drug after you take it.

Pharmacokinetics studies the change in drug concentration in blood over time.

Pharmacokinetics: EFFECT OF THE BODY ON DRUG

Pharmacokinetics



It is often summarised as ADME:

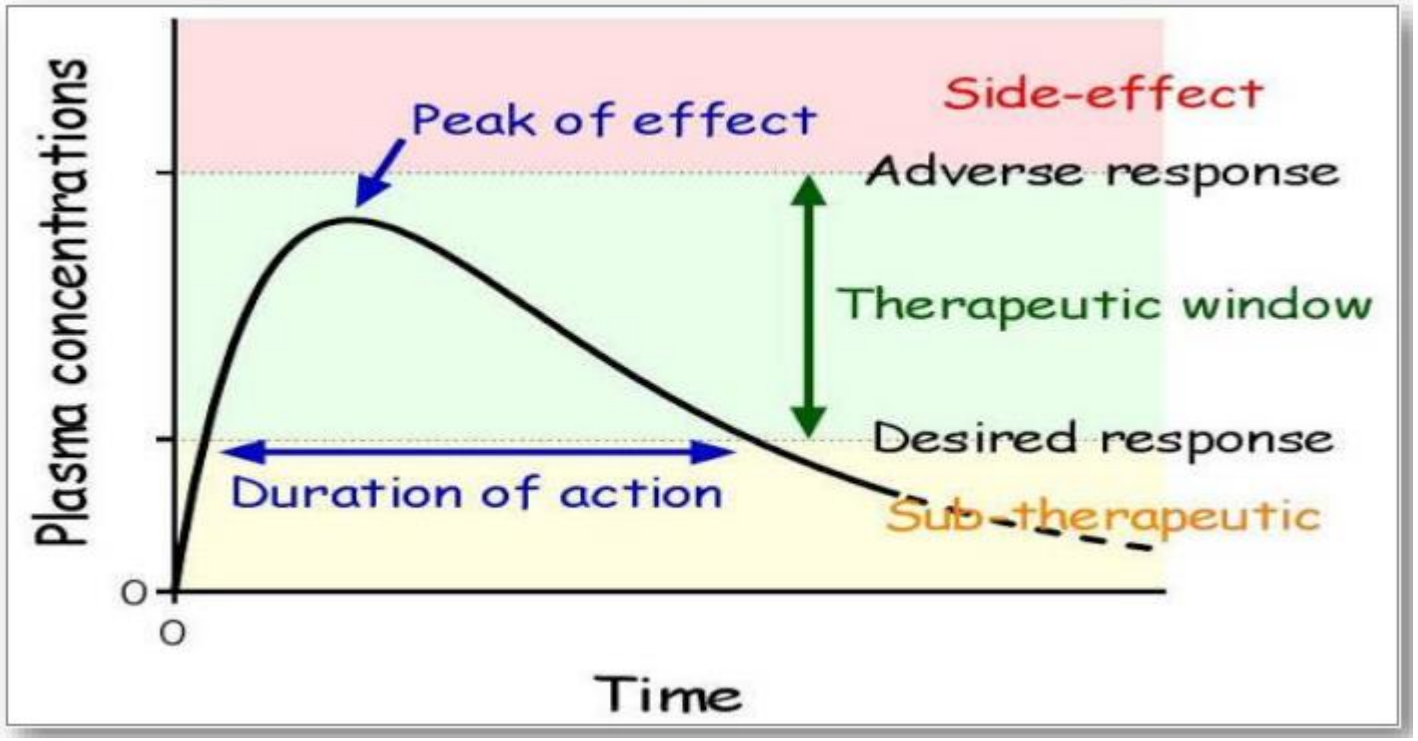
- **A: Absorption** – how the drug moves from **where** you take it (mouth, skin, muscle) into the blood.

- **D: Distribution** – how the drug travels **through** blood to tissues and organs.

- **M: Metabolism** – how the body, mainly the liver, chemically breaks down the drug.

- **E: Excretion** – how the body removes the drug, mainly through the kidneys and urine.

The therapeutic Window concept. It explains how plasma concentration relates to effect.



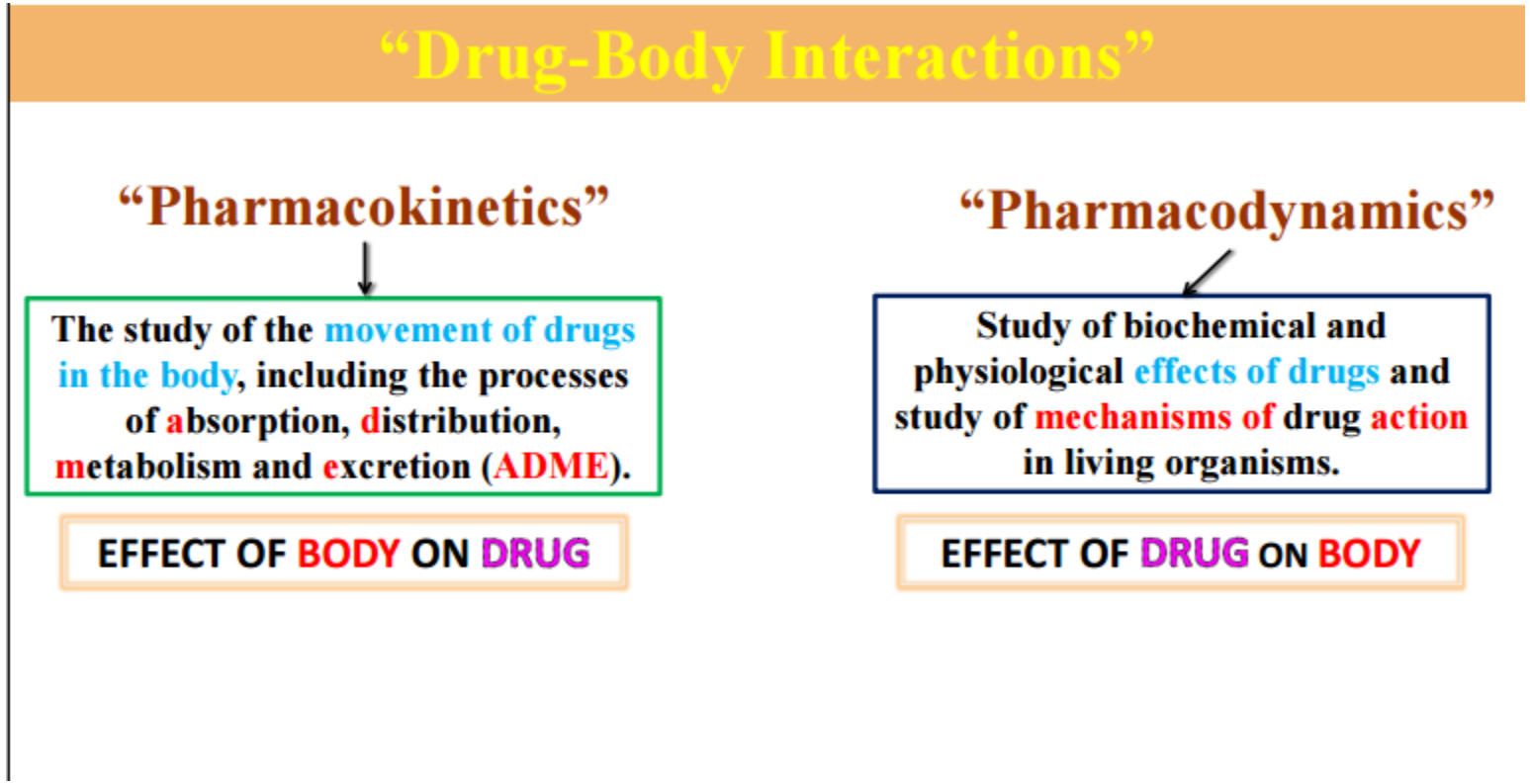
- **Sub-therapeutic zone:** Drug concentration too low to give **a desired response**. No therapeutic benefit.

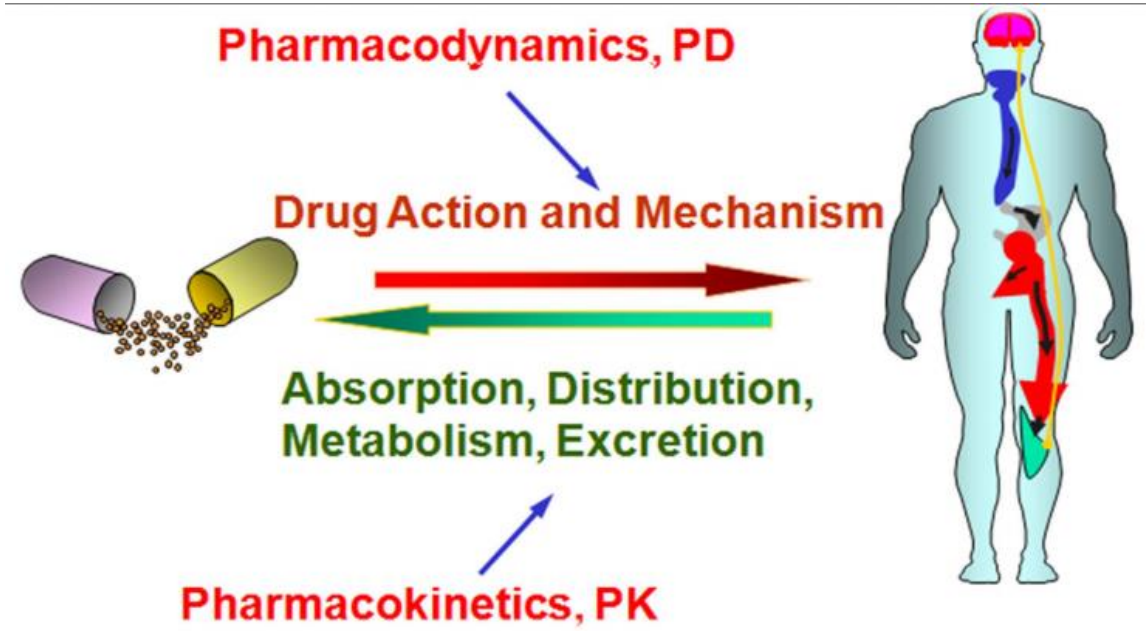
- **Therapeutic window:** This is the target range. Concentration is high enough for **a desired response** but low enough to avoid **an Adverse response**.

- **Adverse response / Side-effect zone:** Concentration too high. Risk of toxicity and side effects.

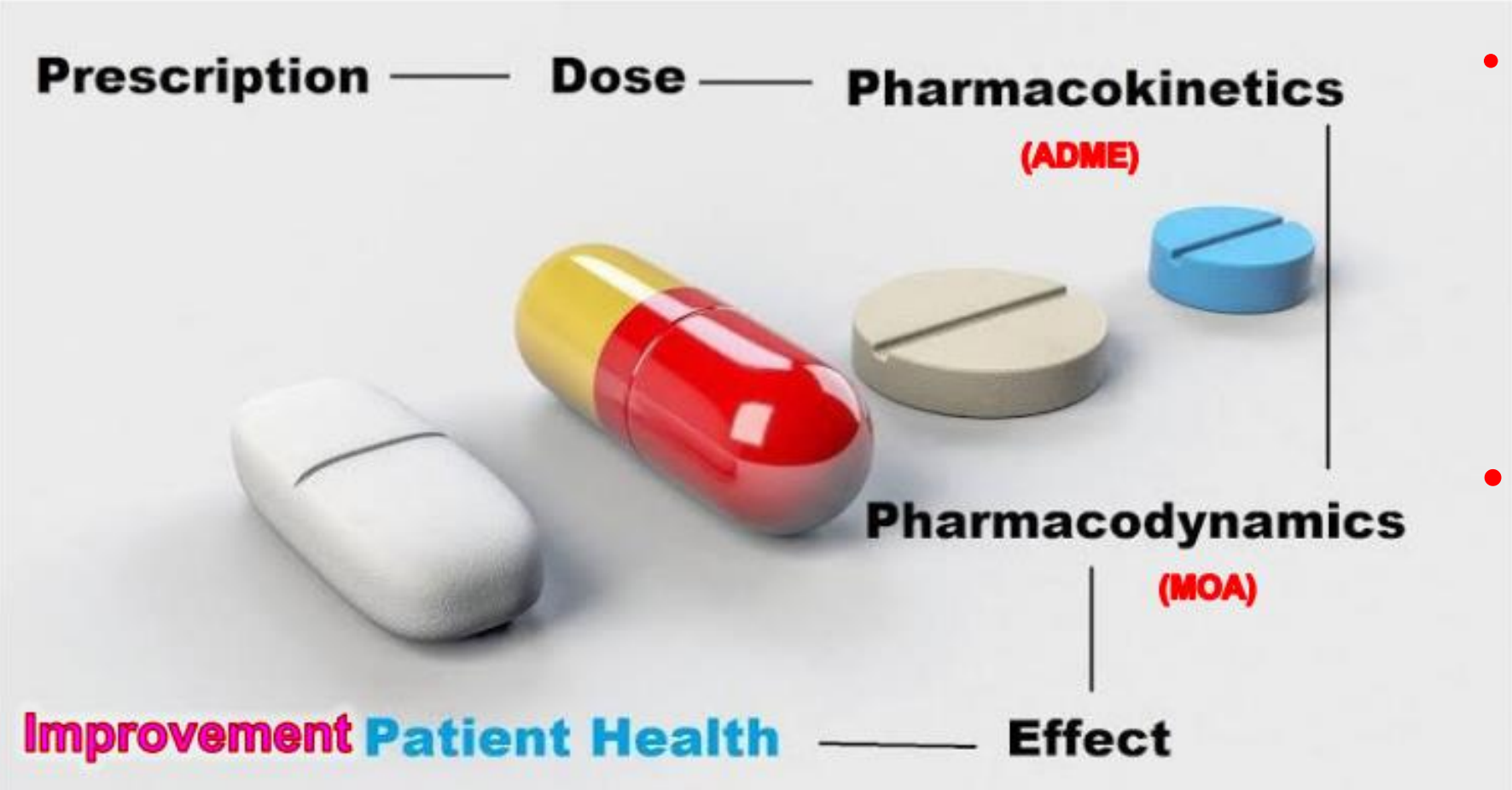
Pharmacodynamics: EFFECT OF DRUG ON THE BODY

- Study of biochemical and physiological effects of drugs and study of mechanisms of drug action in living organisms.





- **Pharmacodynamics, PD: Drug Action and Mechanism**
  - Means: what the drug does to the body.
  - Includes binding to receptors in the brain or organs, changing function, producing therapeutic effects or side effects.
- **Pharmacokinetics, PK: Absorption, Distribution, Metabolism, Excretion**
  - Means: **what the body does to the drug**.
  - Includes how the drug is absorbed in the gut, distributed via blood, metabolised mainly in the liver, and excreted via the kidneys.



- **ADME** (Absorption, Distribution, Metabolism, Excretion) determines how much drug reaches the blood and tissues.
- **MOA (Mechanism of Action)** means how a drug produces its effect, like receptor binding or enzyme blocking.

A = **Absorption**, the first step of ADME.

is the movement of a drug from its site of administration into the bloodstream.

- Without absorption, a drug cannot reach distribution or produce an effect, except for direct IV (Intravenous) administration.

Routes of administration

- **1. Enteral administration:** via gastrointestinal tract
  - **Oral:** swallowed, absorbed in stomach/intestine
  - **Sublingual:** under tongue, fast absorption
  - **Rectal:** via rectum
- **2. Parenteral administration:** bypass gut, injected
  - **Parenteral: IV, IM, SC** as labelled in diagram. IV (Intravenous) is directly into vein, IM (Intramuscular) into muscle, SC (Subcutaneous) under skin**. IV has 100% absorption; no absorption phase needed.**
- **3. Other routes of administration:**
  - **Inhalation:** lungs, rapid, shown in diagram
  - **Transdermal patch:** through skin
  - **Topical:** on skin or mucosa, local effect

Factors influencing absorption of the drug

A. Factors related to the drug:

- **1) Molecular weight of the drug:**
  - Smaller molecules absorb faster and more completely through membranes than larger ones.
  - Large molecular weight drugs have poor diffusion.
- **2) Lipid solubility of the drug:**
  - Cell membranes are lipid. **Lipid-soluble drugs cross easily by passive diffusion**.
  - Water-soluble drugs need special transport and are absorbed more slowly.
- **3) Drug formulation:**
  - How drug is made: solution, tablet, capsule, enteric coated.
  - **Enteric coating prevents breakdown of tablets by the acid pH of the stomach,** allowing release in the alkaline pH of the intestine, where absorption is better. Also changes onset time.
- **4) Drug combination and interactions with food and other drugs influence absorption:**
  - Food can increase or decrease absorption.
  - Example: tetracycline with milk (calcium) decreases absorption; some drugs need food to absorb better.
  - Other drugs can bind the drug in the gut or change gut pH and motility.

B. Factors related to the body.

1) Routes of administration (ROA):

- IV is fastest, 100% absorption. Oral is slower, needs to survive stomach acid and first-pass metabolism.

2) Total surface area available for absorption:

- Larger surface = more absorption. Small intestine has huge surface due to villi, so most drugs absorb there, not in the stomach.

3) Blood flow to absorption site:

- More blood flow maintains concentration gradient and removes drug quickly, increasing absorption. Example: muscle with high blood flow after exercise absorbs IM injection faster.

4) Presence of co-factor:

- Some drugs need carriers or co-factors. Like iron needs vitamin C, or lipid-soluble vitamins need bile salts.
- The bubble-like Dissolves Like means lipid-soluble drugs need a lipid environment; water-soluble drugs need water to dissolve first.

5) Effect of pH on drug absorption:

- As the previous slide explained. Weak acids absorb in acidic stomach pH 1-2 when unionised. Weak bases absorb in an alkaline intestine.

6) Contact time at the absorption surface:

- Longer contact = more absorption. If gut motility is fast, as in diarrhoea, the drug has less time to be absorbed.

7) Gastric emptying time and passage of drug to the intestine may be influenced by gastric contents and intestinal motility:

- Food delays gastric emptying, so the drug stays longer in the stomach, slower to reach the intestine where most absorption happens.
- Motility changes: high motility moves drug fast, low motility delays but may increase contact time.

(A)- absorption

Diffusion of drugs is a key rule for absorption and excretion in the kidney.

Core principle:

- **Unionized drug can diffuse to biologic membranes.**
  - Only non-charged form crosses lipid membrane. Ionized form cannot.

- **Lipid-soluble drugs and weak acids, which are unionized at the low pH (1-2) of gastric contents, may be absorbed directly from the stomach (Vice versa).**

  - Stomach HCl is acidic, with a pH of 1-2. Weak acids stay unionised there, so they are absorbed from the stomach.
  - Weak bases become ionised in the stomach, so they are absorbed better in the intestine, where the pH is alkaline.

Application to kidney elimination:

- **Only non-ionised forms undergo active or passive reabsorption. Ionised forms of drugs are trapped in the filtrate.**

  - In kidney tubule, if drug is ionized in urine, it stays in urine and is excreted. If unionised, it is reabsorbed back to blood.

- **Acidification of urine → increases ionization of weak bases → increases renal elimination.**
  - Make urine acidic, weak base drugs become ionized, trapped in urine, more excreted.
- **Alkalinization of urine → increases ionization of weak acids → increases renal elimination.**
  - Make urine alkaline ; weak acid drugs become ionised, trapped in urine, and more are excreted.

(A)- absorption

Bioavailability of a drug (BA = %)
In points:

- **Definition:** It is the fraction, percentage of the administered dose that **reaches the systemic circulation in active form.**
  - BA (Bioavailability) measures how much drug actually gets to blood to work.
- **The IV Bioavailability equals 100%**
  - IV (Intravenous) goes directly into blood, no loss, so reference standard is 100%.
- **Expressed as % of the dose**
  - Calculated as: (Amount reaching plasma / Dose given) x 100
- **Example given:**
  - Oral dose = 50mg results in 10mg entering plasma. 40mg has been metabolized or lost in gut, liver, etc.
  - BA = 10/50 = 20%

Why it is low for oral: Due to incomplete absorption, acid destruction in stomach, first pass metabolism in liver, and interaction with food.
Clinical meaning: If BA is 20%, you need higher oral dose to get same effect as IV.